Practice — The Heating Effect of Electric Current

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ **[1]**

Heat is produced in a wire because the current faces —

(a) no opposition (b) resistance (c) magnetism (d) light

Q2 · MCQ **[1]**

Nichrome wire is used in heaters because it —

(a) is an insulator (b) generates more heat for a given current (c) is the cheapest metal
(d) never gets hot

Q3 · MCQ **[1]**

Which appliance does NOT mainly use the heating effect of current?

(a) electric iron (b) electric kettle (c) electric fan (d) room heater

Q4 · TRUE / FALSE **[1]**

For the same time, a wire heats up more with two cells than with one cell.

(a) True (b) False

Q5 · ASSERTION–REASON **[1]**

A: An incandescent lamp glows when switched on.
R: Its filament is heated by the current (heating effect).

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK [1]

The coil or rod of wire in a heating appliance is called the heating ________.

____

Q7 · FILL IN THE BLANK [1]

For connecting wires we use ________, which has low resistance (a metal).

____

Short answer (2 marks each)

Q8 [2]

What is the heating effect of electric current? Name two appliances based on it.

Q9 [2]

List three factors on which the heat produced in a wire depends.

Q10 [2]

Why is copper used for connecting wires but nichrome for heating elements?

Long / application (3 marks each)

Q11 **[3]**

A nichrome wire becomes warm AND a compass below it is deflected when current flows. Which effect(s) of current are shown? Explain each.

Q12 **[3]**

Give reasons why electric heating devices (heater, stove) are often more convenient than burning firewood or charcoal, considering society and safety.

Check yourself → Answer key